

Reviewer Pack — Minimal Order of Quasi-Plurality and Resolution Proof Width ...

Entry 80eb801ec897 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Quasi-Plurality and Resolution Proof Width Bound

Entry ID: 80eb801ec897 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-08 03:45:09 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (Quasi-plurality) **Field B** (complexity object): Boolean Satisfiability (Resolution Proof Complexity)

Statement:

For every Boolean satisfiability instance φ , the minimum order of quasi-plurality among all its clauses is linearly correlated with its resolution proof width $w(\varphi)$, such that $\min_order_quasi_plurality(\varphi) = \Theta(w(\varphi))$.

Rationale (proposer's reasoning):

Quasi-plurality measures the complexity of a set of algebraic equations, which can be related to the structure of Boolean formulas. This invariant has not been extensively explored in complexity theory and may reveal new insights into proof complexity.

Taxonomy category: Algebraic Topology × Boolean Satisfiability (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 827b0d3b5466008b

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between $\min_order_quasi_plurality$ and resolution proof width for at least 30 random CNF formulas is ≥ 0.9 , with p-value ≤ 0.01 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	UNCERTAIN	1.00	HITS	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'algebraic topology' AND 'quasi-plurality' AND 'resolution proof complexity' - 'Boolean satisfiability' AND 'resolution proof width' AND 'min_order_quasi_plurality' - linear correlation 'quasi-plurality' 'resolution proof width'

Top relevant hits considered: - [http://arxiv.org/abs/1103.5740v2] Generating and Searching Families of FFT Algorithms - [http://arxiv.org/abs/2206.00903v2] Satisfiability of Quantified Boolean Announcements - [http://arxiv.org/abs/0903.3848v1] Join-irreducible Boolean functions

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.8s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2**n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            if any(abs(c) == abs(clause[0]) for c in clause[1:]):
                continue
            clauses.append(clause)
        return clauses

    def min_order_quasi_plurality(clauses):
        orders = [sum(1 for c in clause if abs(c) == i) for i in range(1, len(clauses[0]) + 1)]
        return max(orders)

    def resolution_width(clauses):
        # Simplified version of resolution width calculation
        # This is a placeholder and should be replaced with actual implementation
        return len(clauses)

    n = random.randint(5, 40)
    clauses = generate_cnf(n)
    min_order = min_order_quasi_plurality(clauses)
    width = resolution_width(clauses)

    return {
        "metric_name": "min_order_quasi_plurality",
```

```

        "metric_value": min_order,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [random.randint(2, 97) for _ in range(30)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it did not complete within the allotted time limit. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 13

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12934
2	propose	ollama_remote	glm4:latest	0	0	11538
3	propose	ollama_remote	glm4:latest	0	0	13097
4	propose	ollama_remote	glm4:latest	0	0	12609
5	propose	ollama_remote	glm4:latest	0	0	11475
6	preregistration	ollama_remote	glm4:latest	0	0	9209
7	novelty	ollama_remote	glm4:latest	0	0	8602
8	novelty	ollama_remote	glm4:latest	0	0	11557
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18209
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13608
11	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11933
12	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7012
13	judge	ollama_remote	glm4:latest	0	0	103629

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 245412 ms total latency. Provider mix: {'ollama_remote': 13}

(full prompt+response transcripts available in *research/audit/80eb801ec897.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/80eb801ec897.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/80eb801ec897.tar.gz* (if generated)